

SIGNALFORGE — STATEMENT OF WORK (SOW)

Project Summary:

SignalForge is a multi-layer intelligence automation system designed to collect, classify, analyze, and deliver geopolitical, defense, maritime, and satellite-based signals. It transforms raw open-source data into analyst-grade insights, ready to publish posts, and strategic intelligence products. The system is designed for manual posting during the early “Ghost Phase” to maintain human texture and avoid AI-detection.

Core Objectives:

1. Automate ingestion of OSINT feeds (RSS, APIs, scrapers, maritime AIS, aviation, satellite snapshots).
2. Normalize, deduplicate, and store data in a Supabase database.
3. Use AI to convert signals into structured intelligence outputs (X posts, LinkedIn posts, expert replies).
4. Deliver content via Telegram with approval buttons.
5. Store approved posts for manual publishing.
6. Build foundation for future expansion: newsletters, memos, RAG chatbot, AOI alerts, enterprise dashboards, and monetization layers.

Phase 1 Deliverables (Ghost Engine — 0–90 Days):

1. Database Architecture (Supabase)

- Tables: signals, insights, events, event_signals, posts_queue, chunks (pgvector), sources.
- Proper indexing, dedup keys, timestamps, metadata fields.

2. Source Loader

- CSV importer for OSINT sources.
- Normalization of type, domain, region, trust, speed, scraper flags.

3. Signal Collector

- RSS ingestion (news agencies, think tanks).

- HTTP/API ingestion (MoDs, gov press feeds).
- Scraper integration via Apify or custom modules.
- Maritime AIS ingestion (MarineTraffic or equivalent).
- Aviation (ADS-B, NOTAM sources).
- Satellite starter workflow: Sentinel/NASA snapshot pull + metadata + GPT-Vision analysis.
- Dedup and normalization pipeline.

4. Event Fusion Layer

- Clustering signals by region, actors, assets, and time windows.
- Event creation, event membership mapping.
- Rolling summaries.

5. Analyst Composer (AI)

- System Prompt v5.0 + User Prompt v5.0.
- Auto-generation of:
 - * X post (concise, tactical)
 - * LinkedIn post (expanded analysis)
 - * 2–3 replies (≤ 25 words each)
 - * Layer (L0–L4) + confidence + foresight line
- Character limit validator.

6. Telegram Digest & Approval Workflow

- Twice-daily digest of pending insights.
- Inline buttons: Approve_X, Approve_LI, Reject, Save_Newsletter.
- Webhook to update statuses and push approved posts to posts_queue.

7. Manual Posting Helper

- Minimal UI or Telegram-based confirmation flow.

- Allows copy-ready text and “mark posted” logging.

Phase 2 Deliverables (Expansion — 90–180 Days):

1. Newsletter Generator

- Weekly 700–900 word intelligence brief from approved posts.

2. Monthly Strategic Memo

- 1,800–2,500 word deep-dive with frameworks and watch-points.

3. Embeddings & RAG Chatbot

- Nightly chunking + vector embeddings.
- API interface to query archive with citations.

4. Accuracy Tracking System

- Forecast hit/miss tracking.
- Rolling hit rate.
- Warning-time metrics.

Phase 3 Deliverables (Advanced IMINT — 180–365 Days):

1. High-Resolution Satellite Integration

- Integration with Planet Labs, Capella, or equivalent.
- Automated AOI snapshots + OpenCV change detection.

2. AOI Alerts

- Automated anomaly detection feeding Telegram alerts.

Phase 4 Deliverables (Revenue Layer):

1. Paid newsletter.

2. Monthly and quarterly strategic briefs for institutions.
3. Custom intelligence reports.
4. Private briefings and webinars.
5. Enterprise API access.
6. Fused dashboards (OSINT + IMINT + maritime + aviation).

Tech Stack Requirements:

- Automations: n8n, Apify (optional), Node/FastAPI for custom endpoints.
- Database: Supabase + pgvector.
- AI: GPT-5 for insights; GPT-4o/mini for preprocessing; Vision models for IMINT.
- Hosting: Supabase + Render/Hostinger for n8n.
- Frontend (minimal): Internal UI for post queue review.

Development Roadmap:

Phase 1 — Ghost Engine (0–90 days)

Phase 2 — Knowledge & Product Expansion (90–180 days)

Phase 3 — Satellite Fusion & AOI Alerts (180–365 days)

Phase 4 — Monetization & Enterprise Scale

End Goal:

A fully-automated intelligence backbone capable of powering a private think-tank, generating analyst-grade content daily, supporting paid intelligence products, and scaling toward enterprise revenue (\$10k–100k/month).